

NEUROSCIENCE

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Lesson one

2 hours

What is neuroscience.

- **Neuroscience** is the scientific study of the nervous system.
- It is a multidisciplinary branch of biology that combines physiology anatomy etc to understand the fundamental and emergent properties of neurons and neural circuits.

Overview **of the** Nervous System

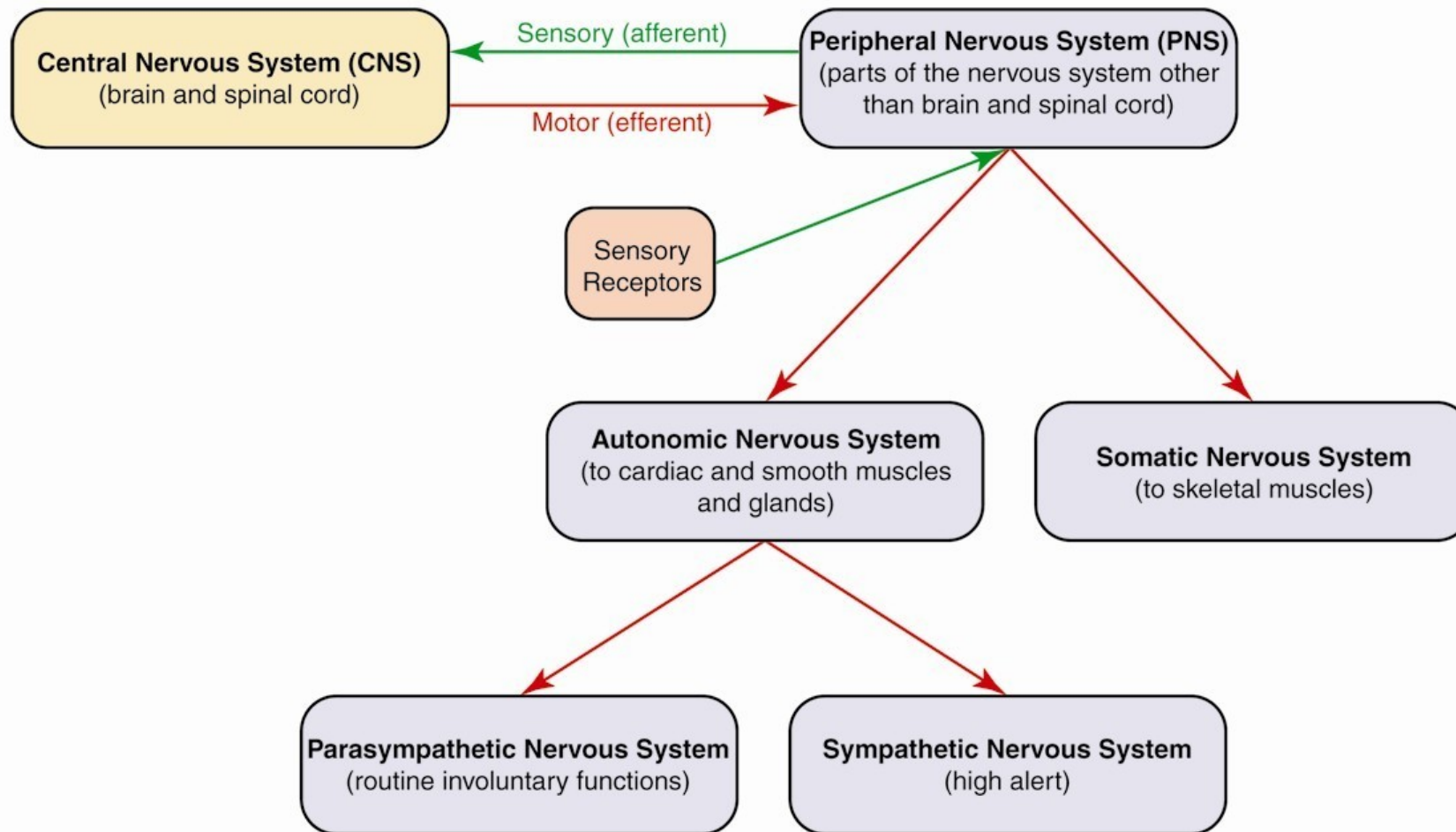
Nervous System Overview

- organization of the nervous system
 - two major divisions
 - the efferent nerves
- nervous tissues
 - neuroglia
 - neurons

Organization of the Nervous System

- two major divisions
 - central nervous system (CNS)
 - peripheral nervous system (PNS)
 - sensory receptors
 - afferent (sensory) nerves
 - efferent (motor) nerves

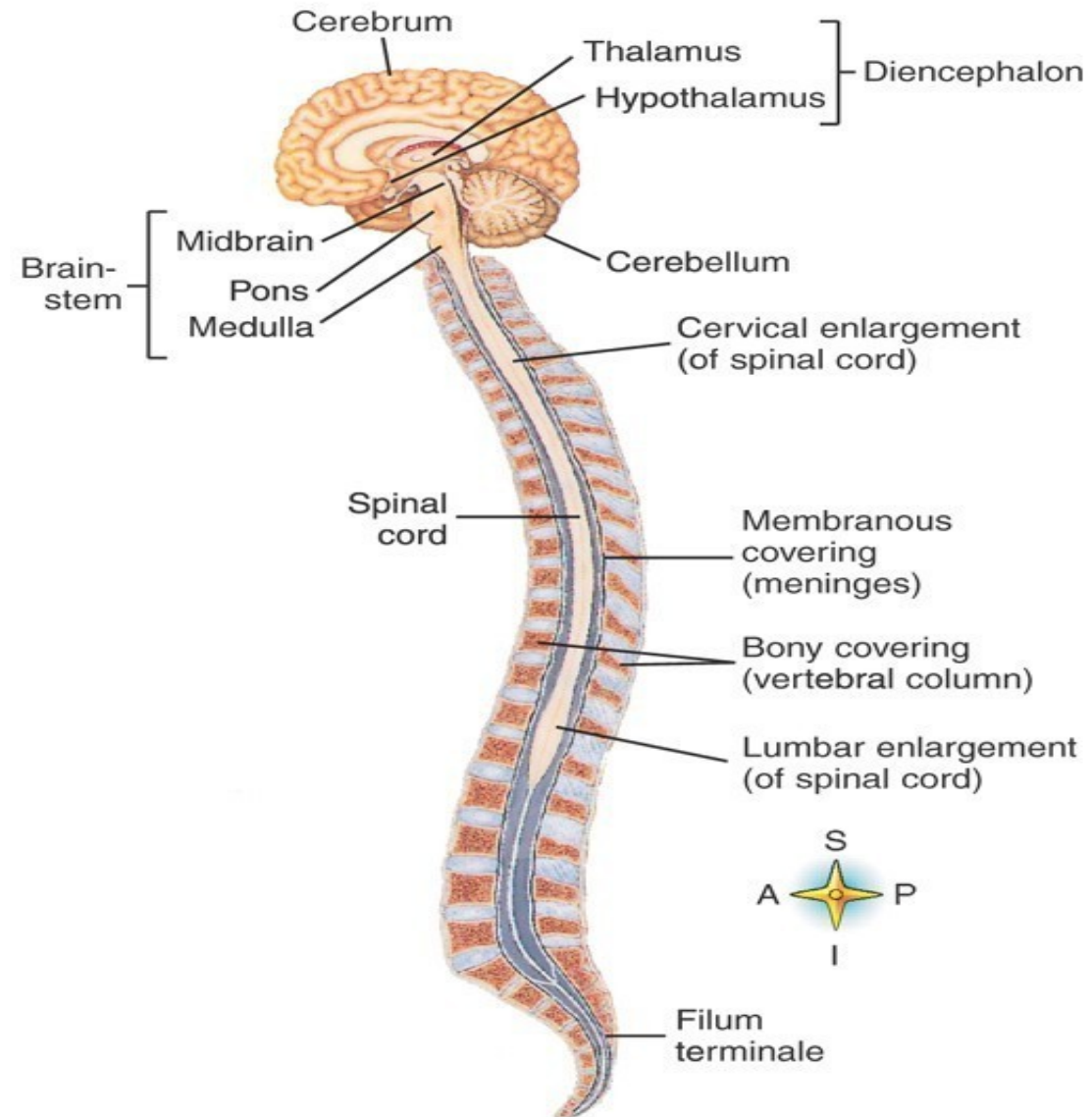
Two Major Divisions



Functional Anatomy of Central Nervous System

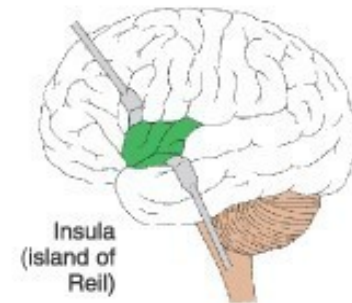
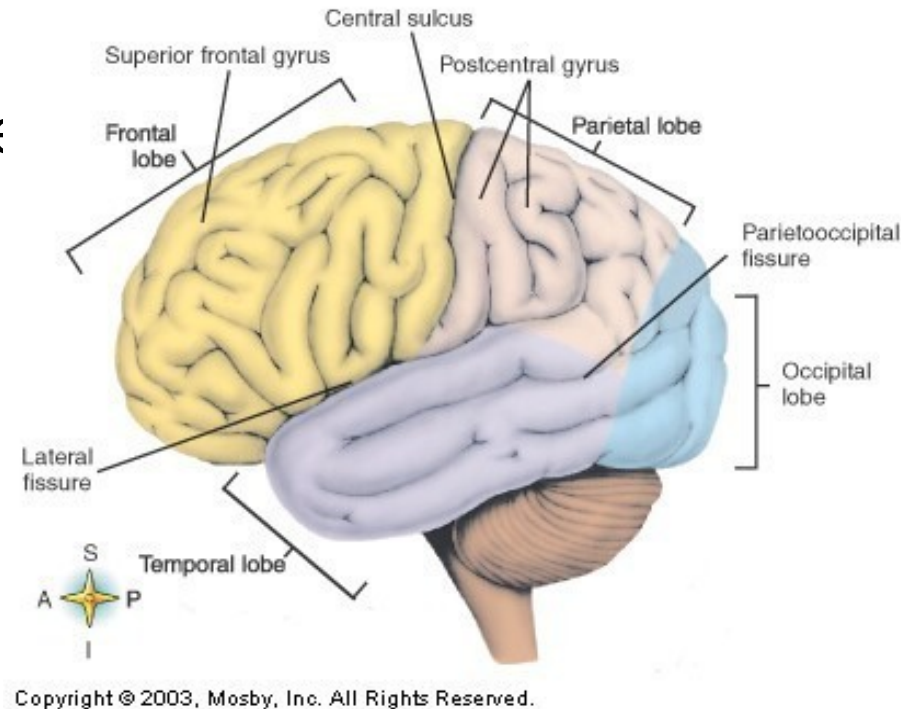
The Brain

- cerebrum
- diencephalon
- brain stem
- cerebellum
- meninges
- blood-brain barrier

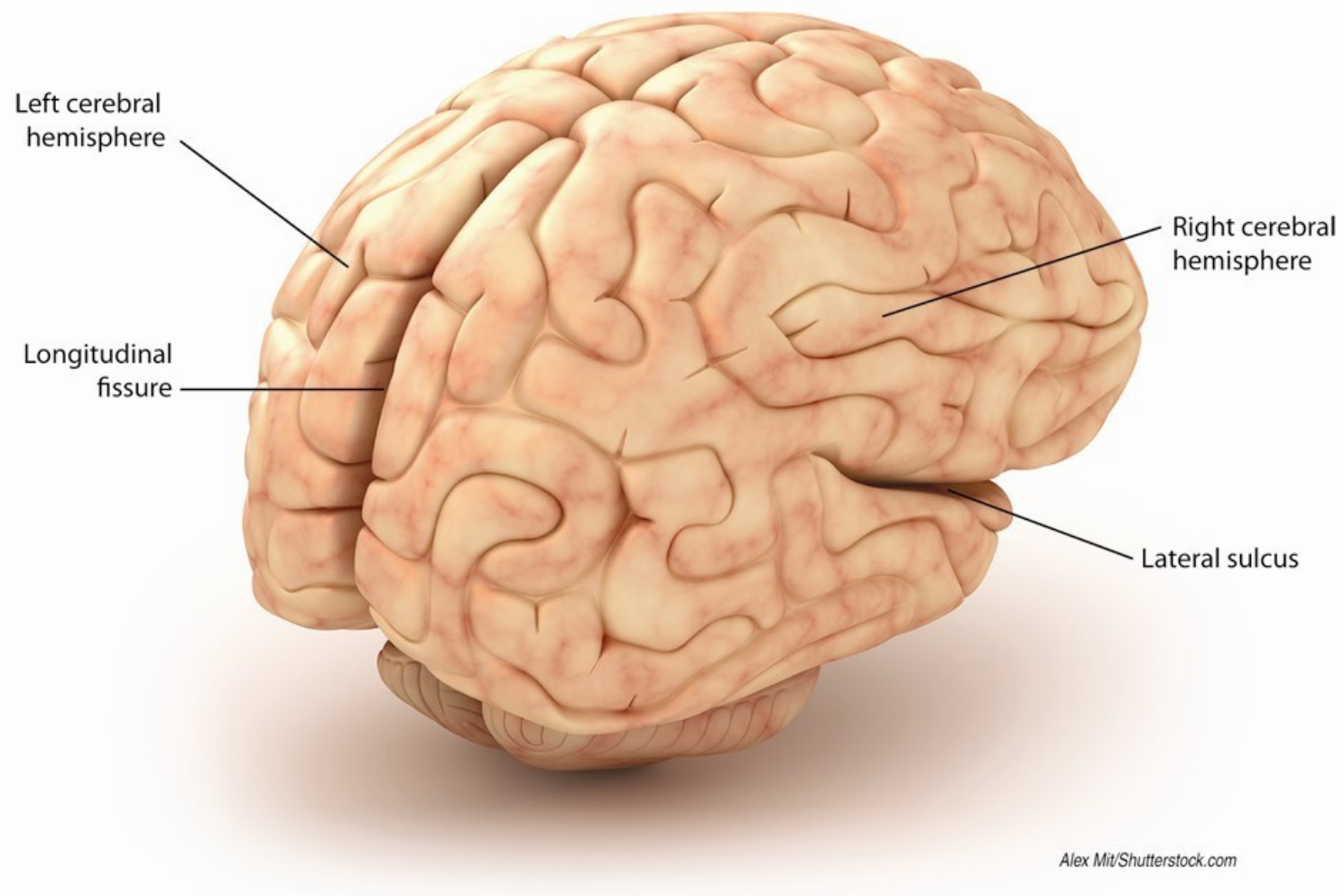


Cerebrum

- cerebral cortex
 - gyrus
 - sulcus
 - fissure
- lobes
 - frontal
 - parietal
 - occipital
 - temporal
- primary motor cortex
- primary somatic sensory cortex



Cerebrum



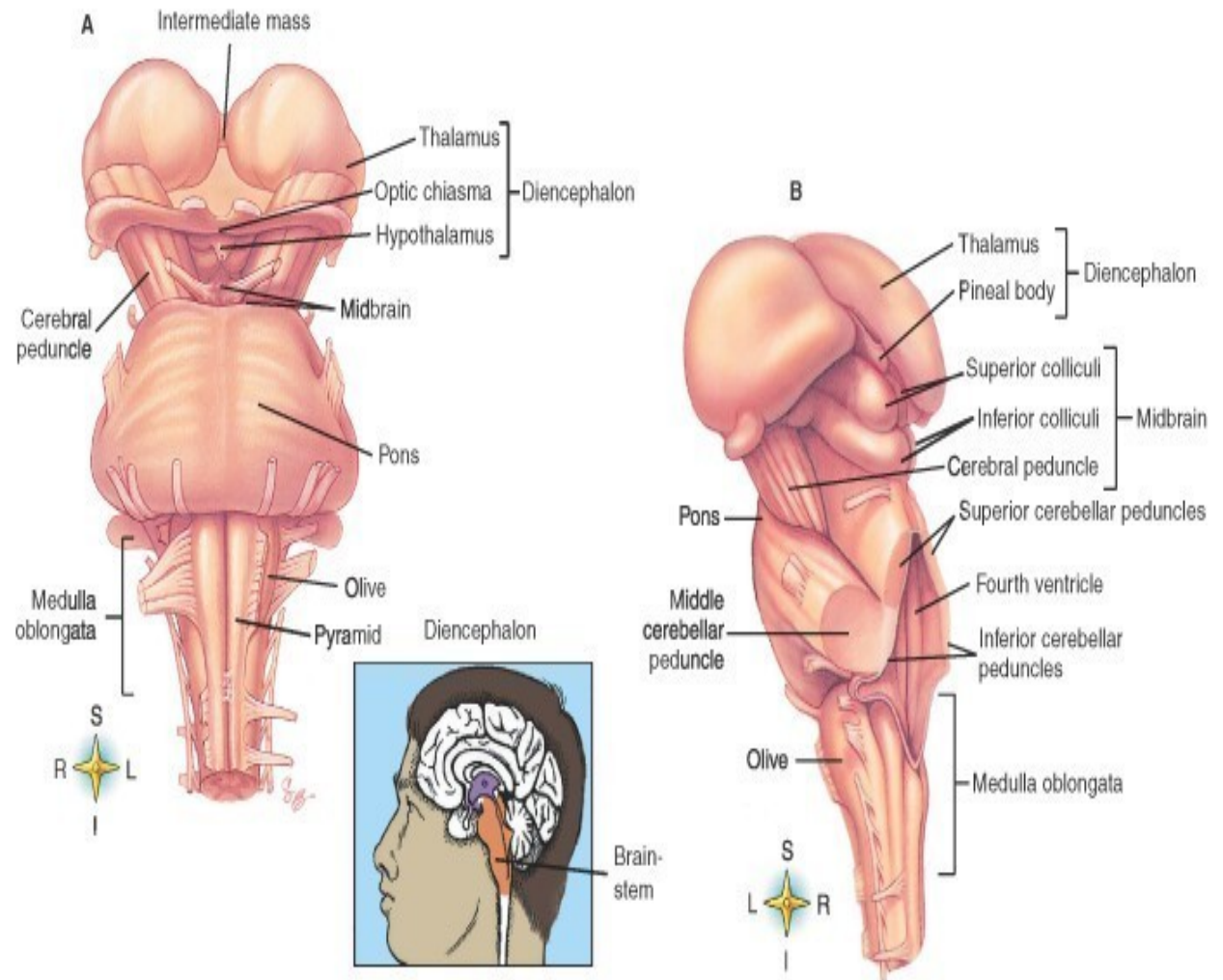
Brain Stem and Diencephalon

Diencephalon

- thalamus
- hypothalamus
- epithalamus

Brain Stem

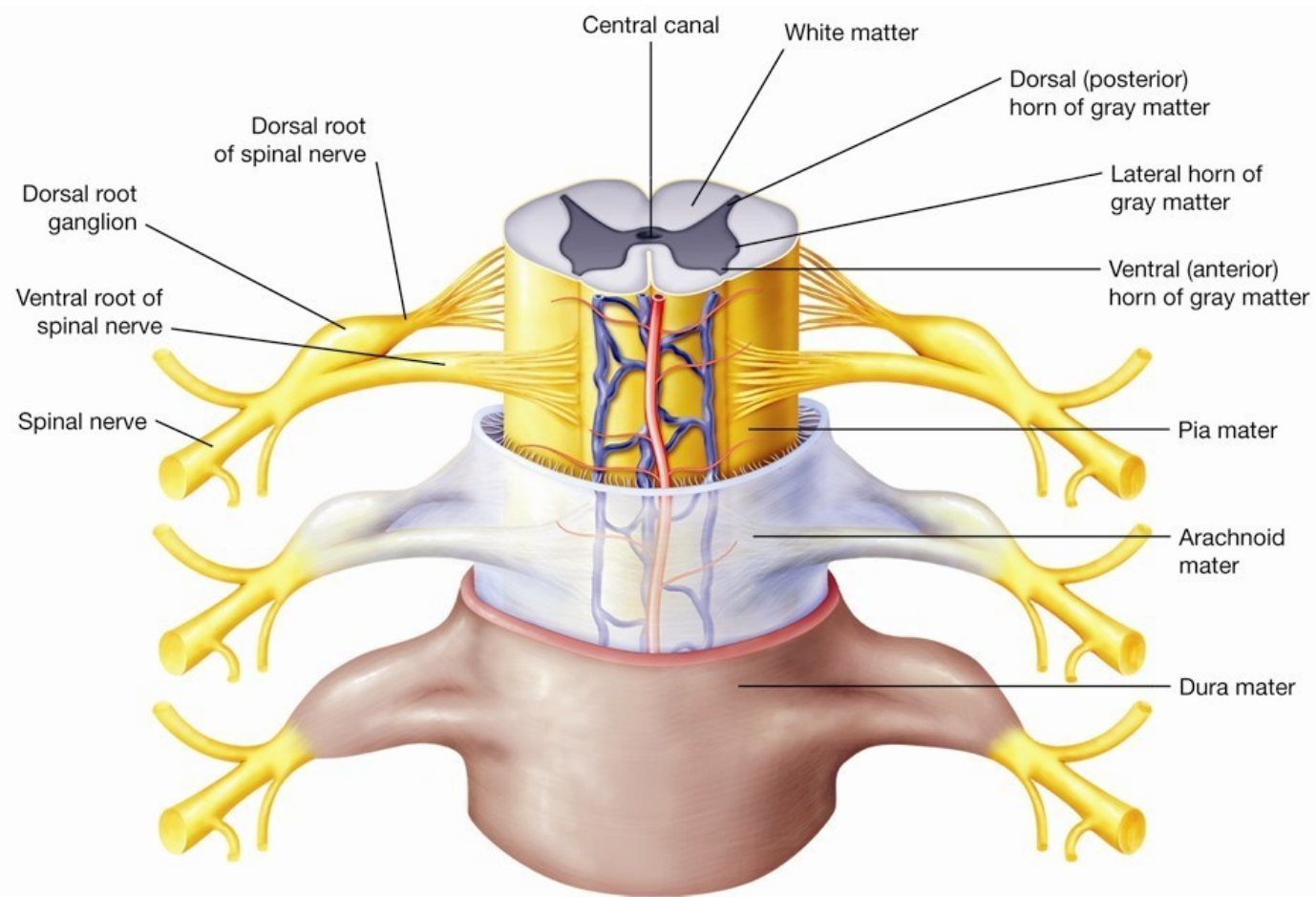
- midbrain
- pons
- medulla oblongata



The Brain

- cerebellum
- blood-brain barrier
- meninges
 - dura mater
 - arachnoid mater
 - pia mater

Spinal Cord

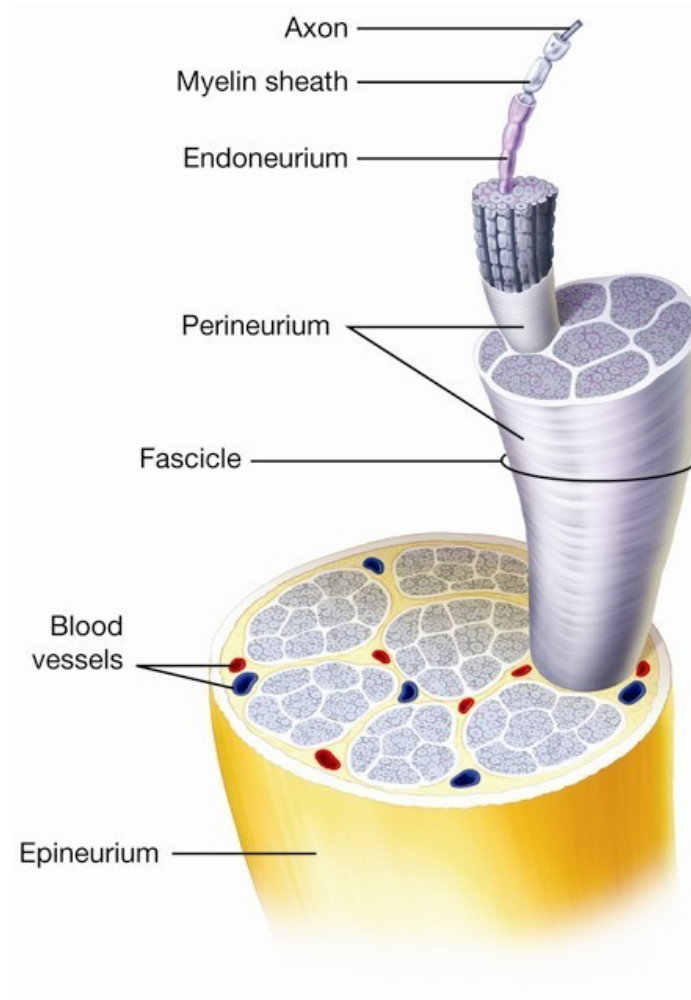


Functional Anatomy of the Peripheral Nervous System

- nerve structure
- cranial nerves
- spinal nerves and nerve plexuses
- autonomic nervous system

Nerve Structure

- endoneurium
 - covers axons
- perineurium
 - bundles fascicles
- epineurium
 - wraps nerves



Autonomic Nervous System

- preganglionic and postganglionic neurons
- sympathetic nerves
 - fight-or-flight action
- parasympathetic nerves
 - resting or digesting action

The Efferent Nerves

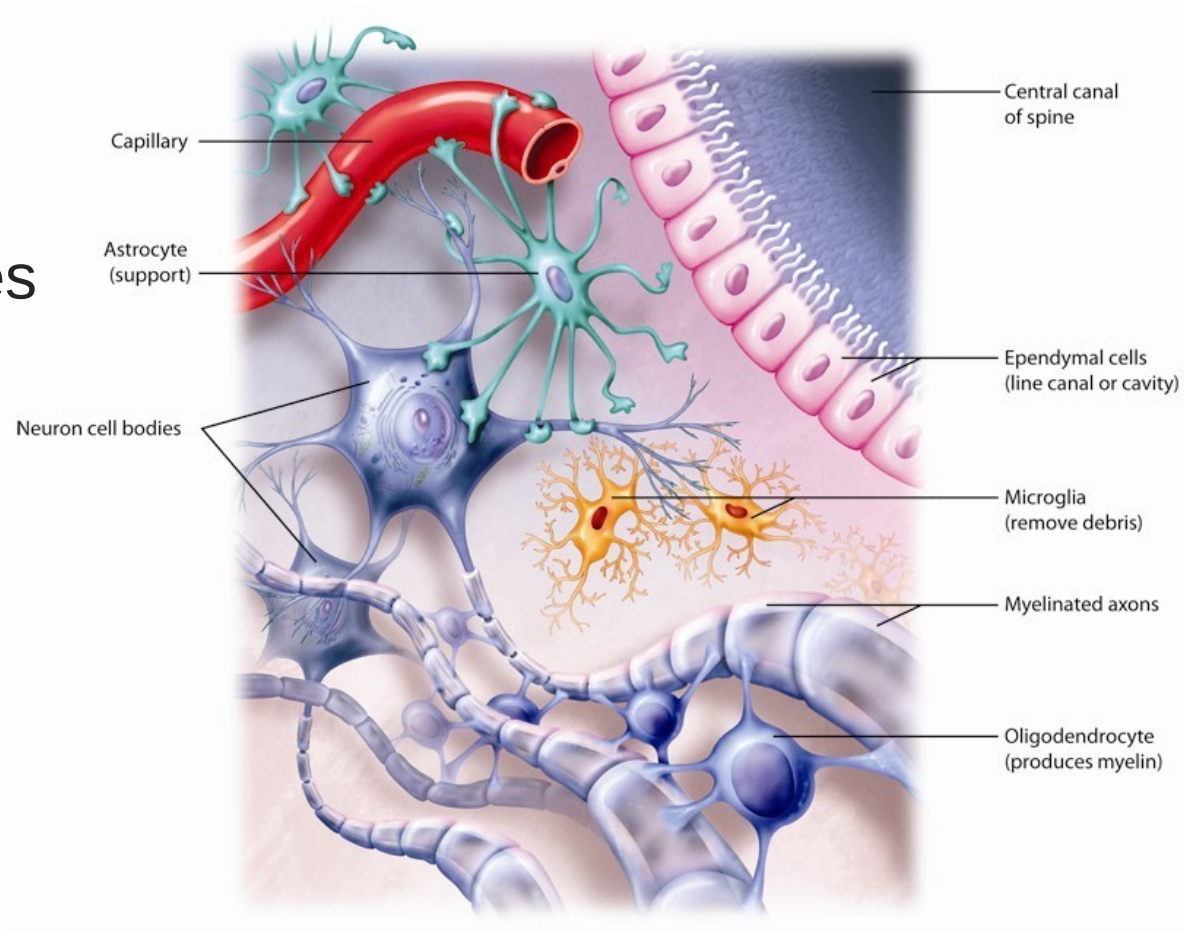
- somatic nervous system
 - voluntary
- autonomic nervous system
 - involuntary
 - Two Divisions
 - sympathetic
 - parasympathetic

Nervous Tissues

- neuroglia
 - also known as *glial cells*
 - support the neurons
 - protect the neurons
 - continue to divide through adulthood- potential consequence is abnormal cell division (cancer)
 - 5 major types of glia cells
- neurons
 - transmit nerve impulses
 - consist of a cell body and at least two processes.

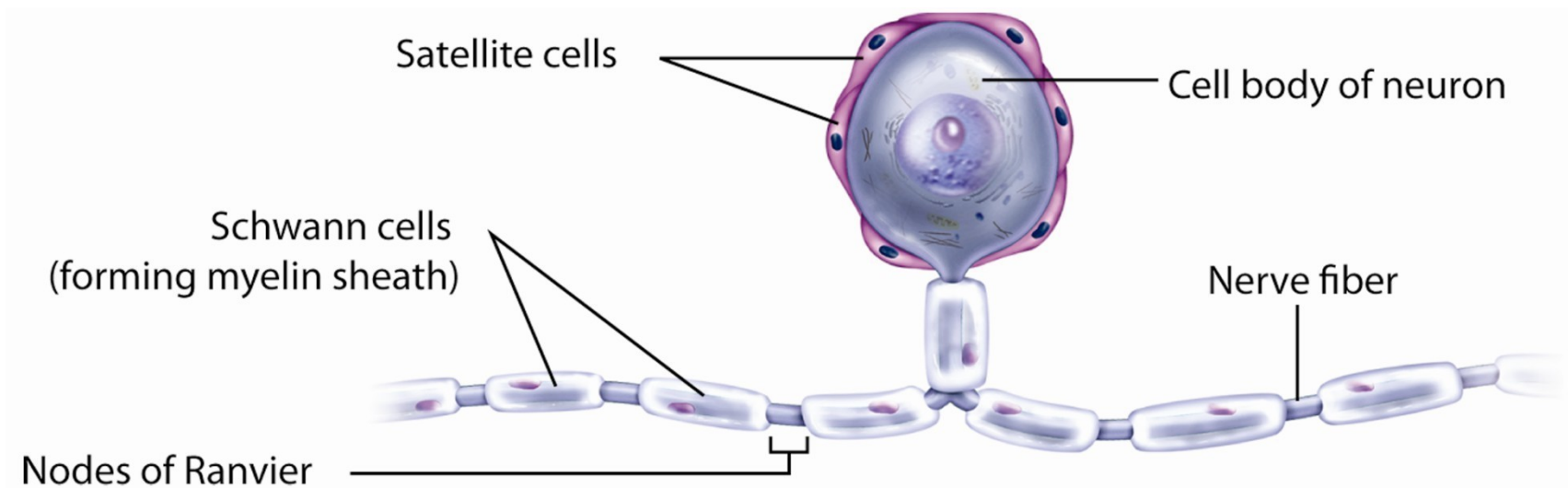
Neuroglia

- central nervous system
 - astrocytes
 - microglia
 - ependymal
 - oligodendrocytes



Neurog lia

- peripheral nervous system
 - Schwann cells
 - satellite cells



Neurons

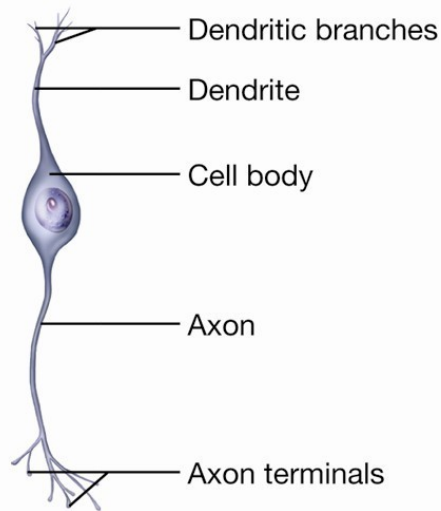
- dendrites
 - send information to cell body
- cell bodies
- axons
 - send information away from cell body

Neuron Types by Function

- sensory neurons
 - send impulses toward CNS
- motor neurons
 - send impulses away from CNS
- Interneurons
 - bridges between neurons

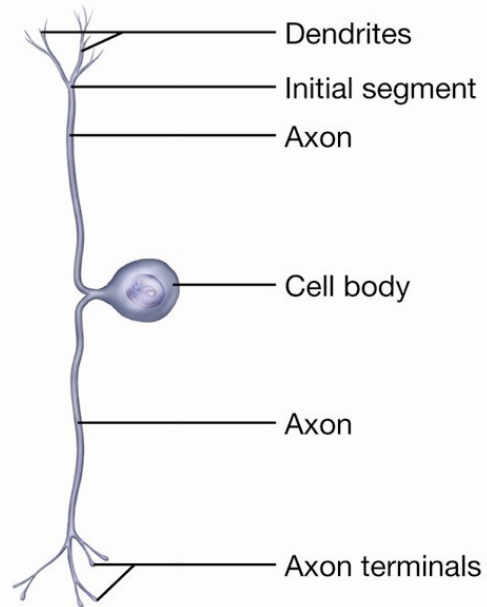
Neuron Structures

- bipolar
 - one axon and one dendrite



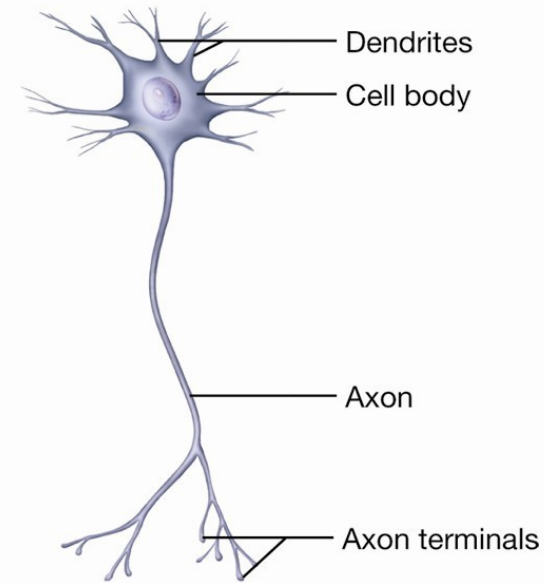
A. Bipolar neuron

- unipolar
 - one axon



B. Unipolar neuron

- multipolar
 - one axon and many dendrites



C. Multipolar neuron

Transmission of Nerve Impulses

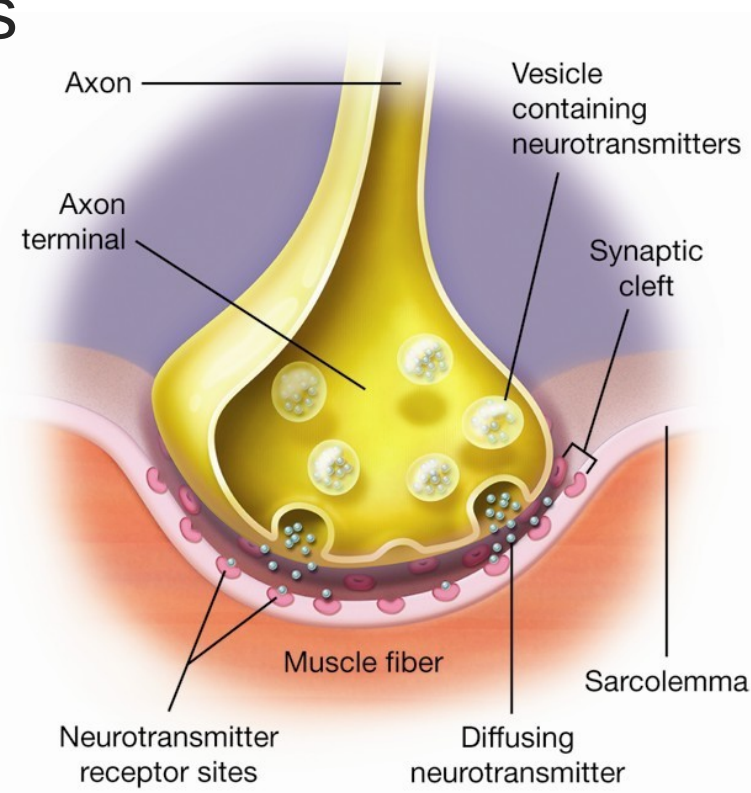
- action potentials
 - change in cell membrane charge
- impulse transmission
 - charge change travels along axon
- reflexes
 - involuntary response to stimulus

Action Potentials

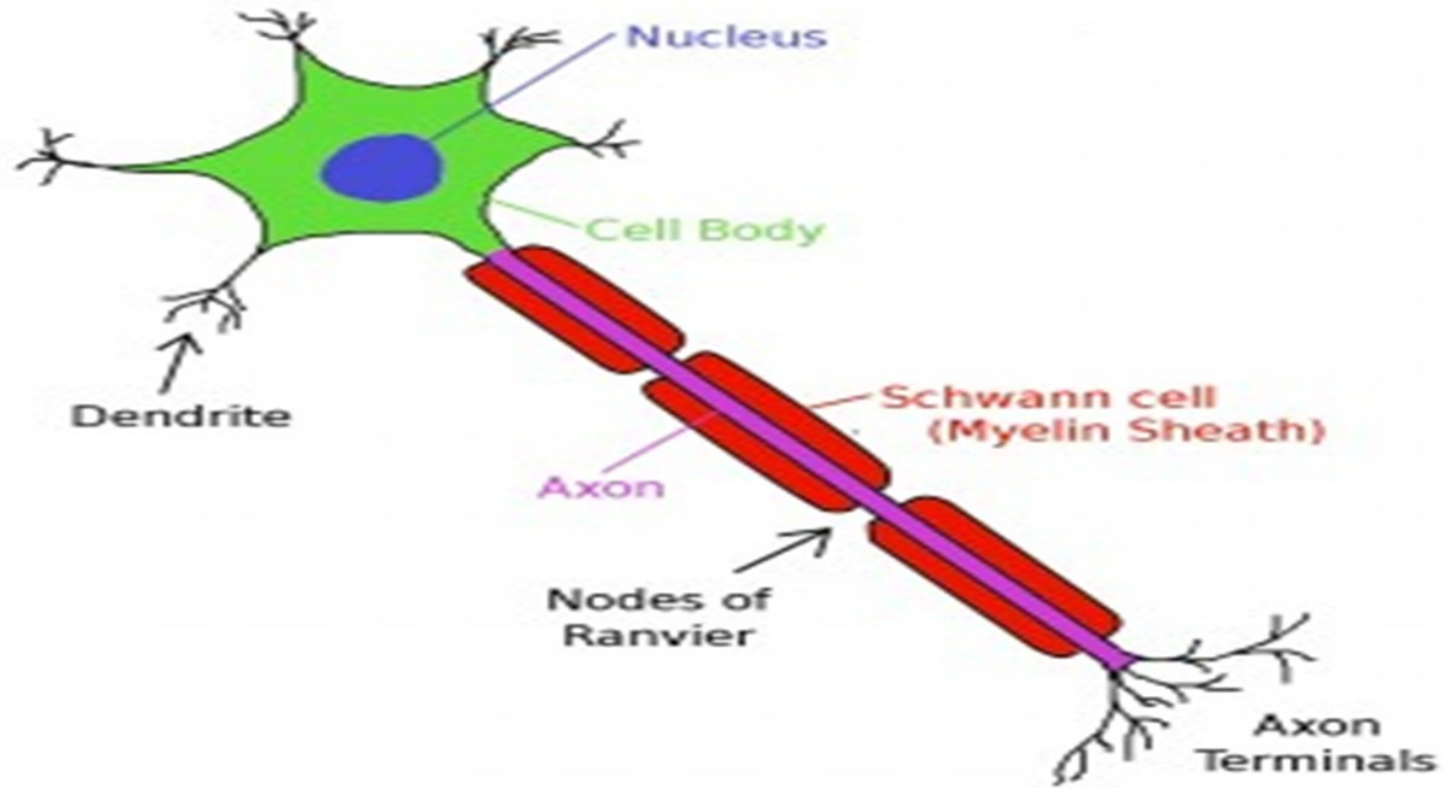
- polarized
- depolarized
- repolarized
- refractory period

Impulse Transmission

- saltatory conduction
- transmission at synapses
 - gap junctions
 - neurotransmitter
 - excitatory effect
 - inhibitory effect



A neuron



Classification of nerve injuries

- **Neurapraxia** – this is a focal nerve compression“ nerve contusion or stretch leading to reversible conduction block.
- **Axonotmesis**
- incomplete nerve injury with axon and myelin sheath disruption but still intact leading to focal conduction block.
- **Neurotmesis**- this is a complete nerve injury with disruption of endoneurium, all connective tissues disrupted, the axon not intact.

Reflexes

- somatic
 - stimulate skeletal muscles
- autonomic
 - stimulate involuntary muscles

Injuries and Disorders of the Nervous System

Disorders of the nervous

- Disorders of the nervous system may involve the following:
- **Vascular disorders**, such as stroke, transient ischemic attack (TIA), subarachnoid hemorrhage, subdural hemorrhage and hematoma, and extradural hemorrhage
- **Infections**, such as meningitis, encephalitis, polio, and epidural abscess
- **Structural disorders**, such as brain or spinal cord injury, Bell's palsy, cervical spondylosis, carpal tunnel syndrome, brain or spinal cord tumors, peripheral neuropathy, and Guillain-Barré syndrome

Cont'd

- Functional disorders, such as headache, epilepsy, dizziness, and neuralgia
- Degeneration, such as Parkinson disease, multiple sclerosis, amyotrophic lateral sclerosis (ALS), Huntington chorea, and Alzheimer disease

General Signs and symptoms of nervous system disorders

- Persistent or sudden onset of a headache
- A headache that changes or is different
- Loss of feeling or tingling
- Weakness or loss of muscle strength
- Loss of sight or double vision
- Memory loss
- Impaired mental ability
- Lack of coordination

Cont'd

- Muscle rigidity
- Tremors and seizures
- Back pain which radiates to the feet, toes, or other parts of the body
- Muscle wasting and slurred speech
- New language impairment (expression or comprehension)

Injuries to the Brain and Spinal Cord

- traumatic brain injury
- cerebral palsy
- spinal cord injury

Traumatic Brain Injury

- violent impact to head
 - mild
 - moderate
 - severe

Cerebral Palsy

- damage to brain
 - before birth
 - during birth
 - during infancy
- motor function impairment

Spinal Cord Injuries

- C1–C3: usually fatal
- C1–C4: quadriplegia
- C5–C7: paralysis of lower extremities
- T1–L5: paraplegia



Corepics/Shutterstock.com

Theoretical models of motor learning and control.

- Motor learning is the understanding of acquisition and/or modification of movement.
- As applied to patients, motor learning involves the reacquisition of previously learned movement skills that are lost due to pathology or sensory, motor, or cognitive impairments. This process is often referred to as recovery of function.

Motor control

- Motor control is how system works. Its all about to activate and coordinate, regulate and direct muscles and limbs involved in the performance of posture or movement.
- Its an interaction of three elements;
- i.e. Task, person and environment.

Objectives:

- Review the principles and foundations of motor learning.
- Compare and contrast motor learning theories.
- Discuss the interaction of the person, task, and environment within the scope of motor control and motor learning.

Characteristics of motor learning include

- A set of internal ***Processes***
- Produces an ***acquired capability*** for movement
- Occurs as a direct result of ***instruction, practice, and experience***
- Is ***NOT directly observable***, and therefore is inferred
- Produces ***relatively permanent changes*** in the capability for skilled movement.

(Magill, 2001; Schmidt, 1991; Rose, 1991)

What is *Learning* vs *Performance*?

5 Characteristics of Learning:

- Performance

- Consistency

Timing
Accurac

- Persistence

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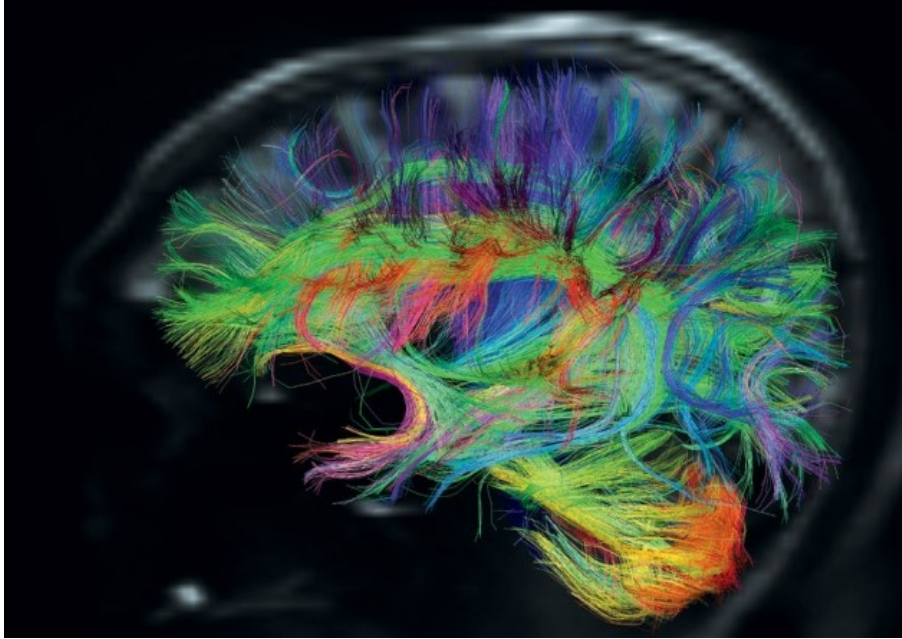
- Stability

- Adaptability

Assessing Motor Learning

- *Performance Curves* – Records levels of performance over time
 - Graph of outcome measures of performance
 - Performance measure plotted on vertical axis
 - Time over which performance is plotted on horizontal axis
- *Retention Test* – A common measure to assess the performance characteristic of the persistence characteristic of improved performance
- *Transfer Test* – Assess the performance characteristics of adaptability by performing a practiced skill in:
 - Novel context (feedback and physical environment changes)
 - Novel skill variations (cup or glass, fast or slow, harder or softer)

Motor Learning vs Neuroplasticity... *What's the Difference?*



Neuroplasticity occurs based on experience – good or bad

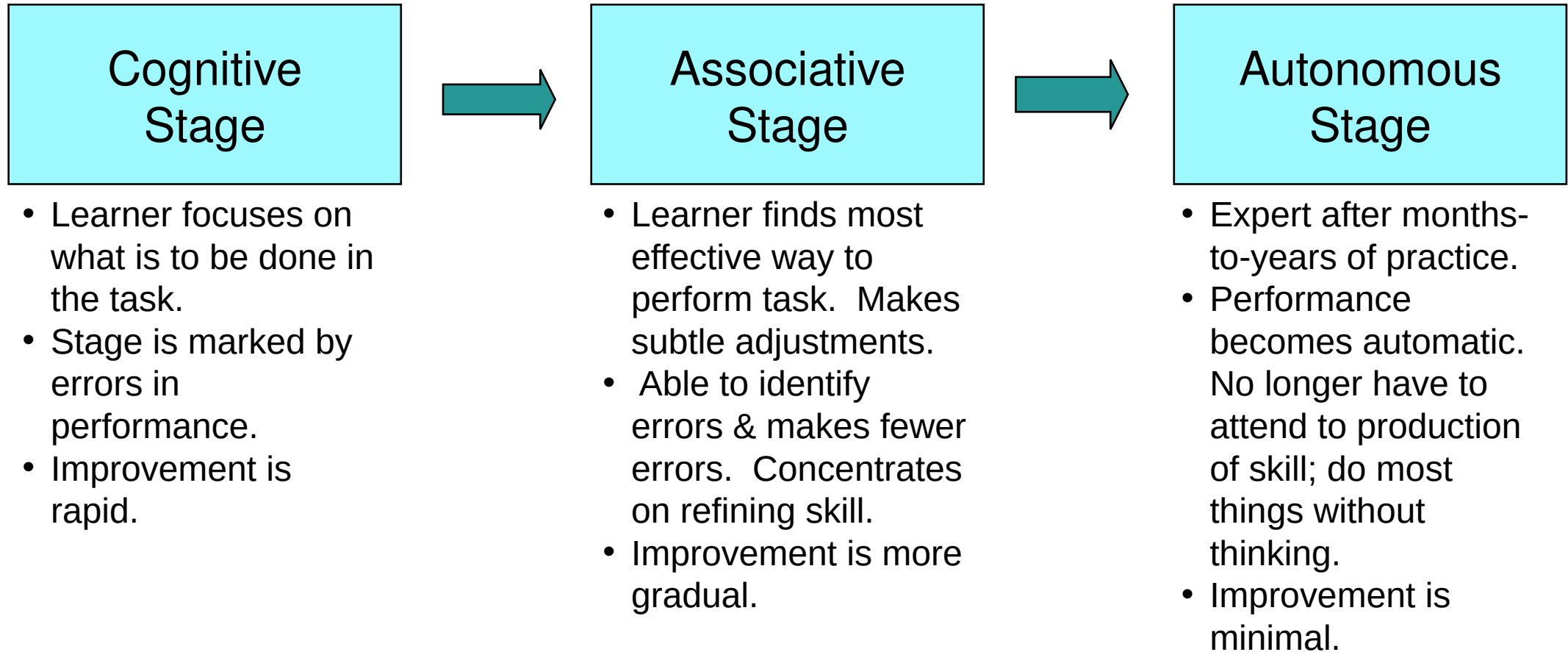


Motor Learning harnesses the power of neuroplasticity to acquire a desired skill

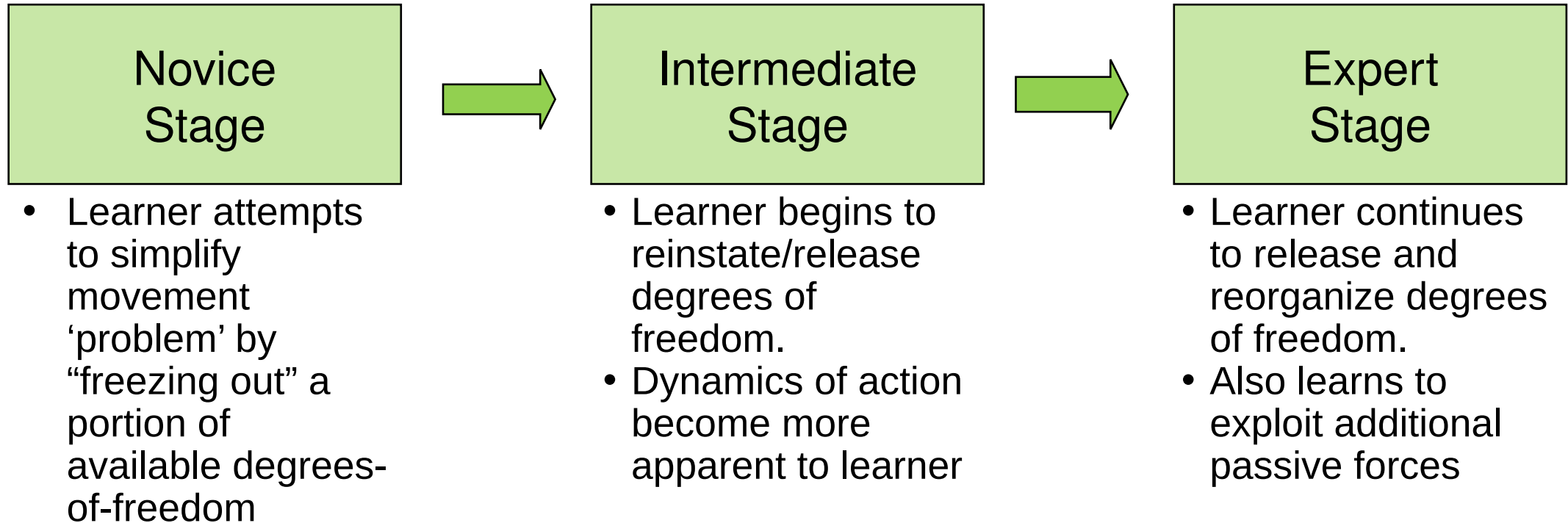
Motor Learning Theories

- 3-Stage Model: Fitts and Posner
- 3-Stage Model: Bernstein
- 2-Stage Model: Gentile

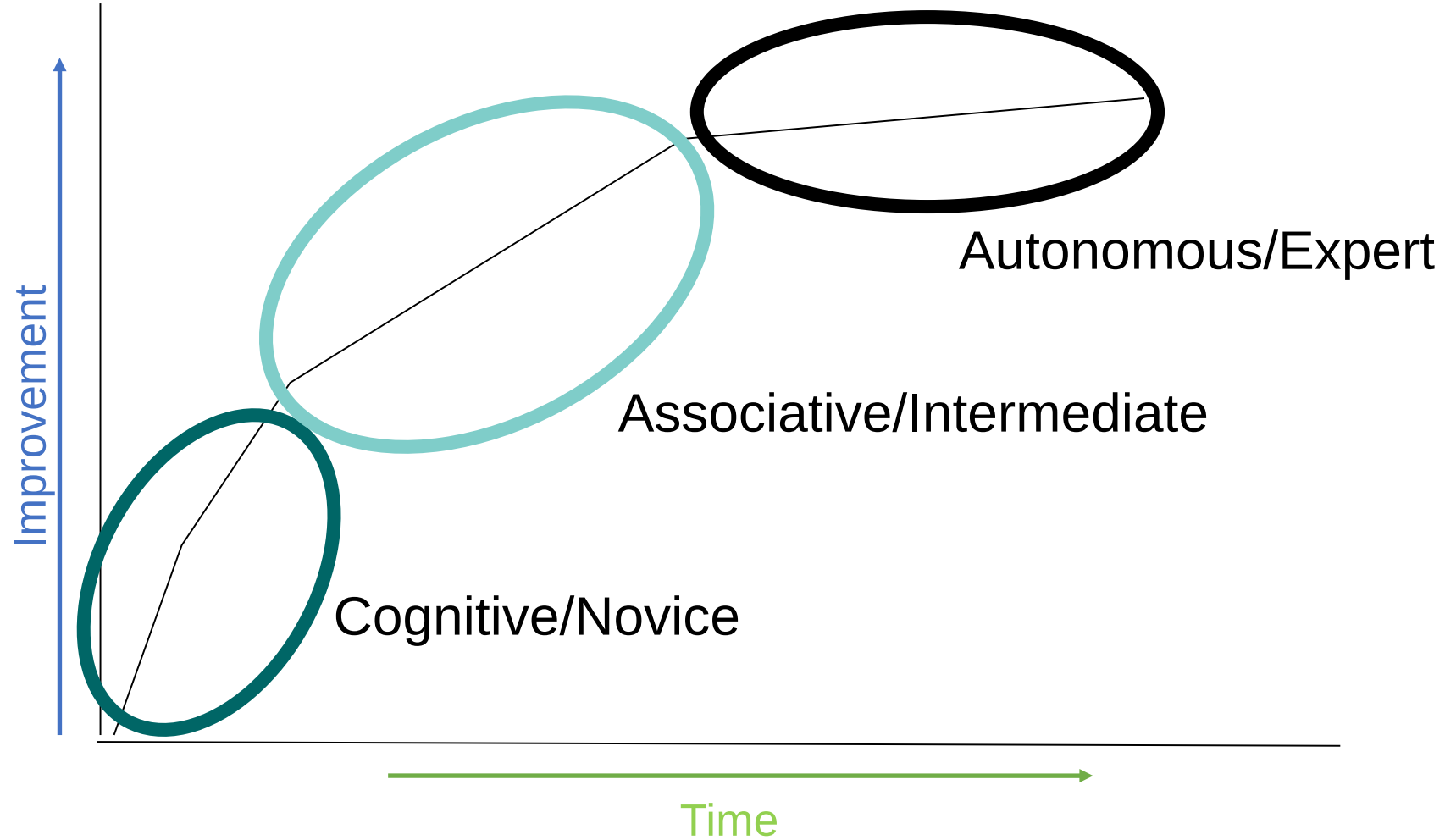
3-Stage Models: Fitts and Posner



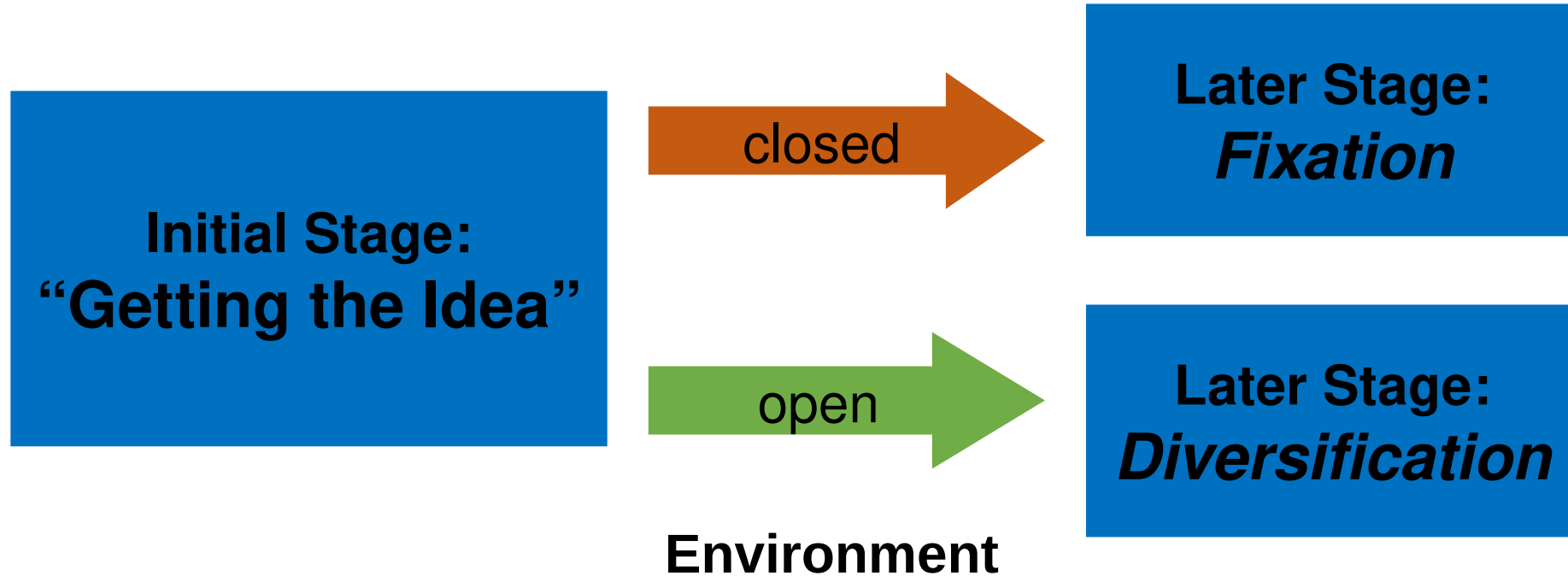
3-Stage Models: Bernstein



Improvement in Motor Learning



Gentile's 2-Stage Model



Gentile's 2-Stage Model

- **“Getting the idea” of the movement**
 - Getting the general concept of what must be done to accomplish the goal of the skill
 - Learner must establish relevant and non-relevant stimuli
 - Learner must establish most appropriate movement pattern to effectively attain the goal of the skill—involves coordination
- **Later Stage: Fixation** (closed skill requirement)
 - Learner must develop capability of doing what is needed-regardless of the situation—adapting
- **Later Stage: Diversification** (open skill performance)
 - Learner must develop a larger repertoire of motor patterns

Gentile's 2-Stage Model

INITIAL STAGE

Getting the



FIXATION

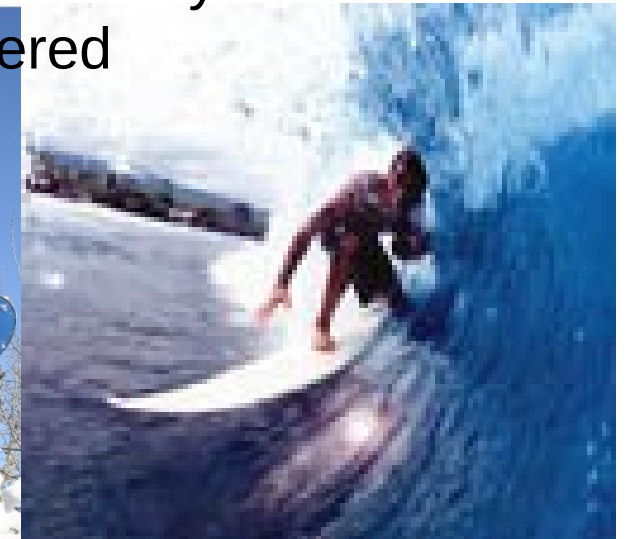
Skills refined and
retained



LATER STAGES

DIVERSIFICATION

Skills markedly
altered

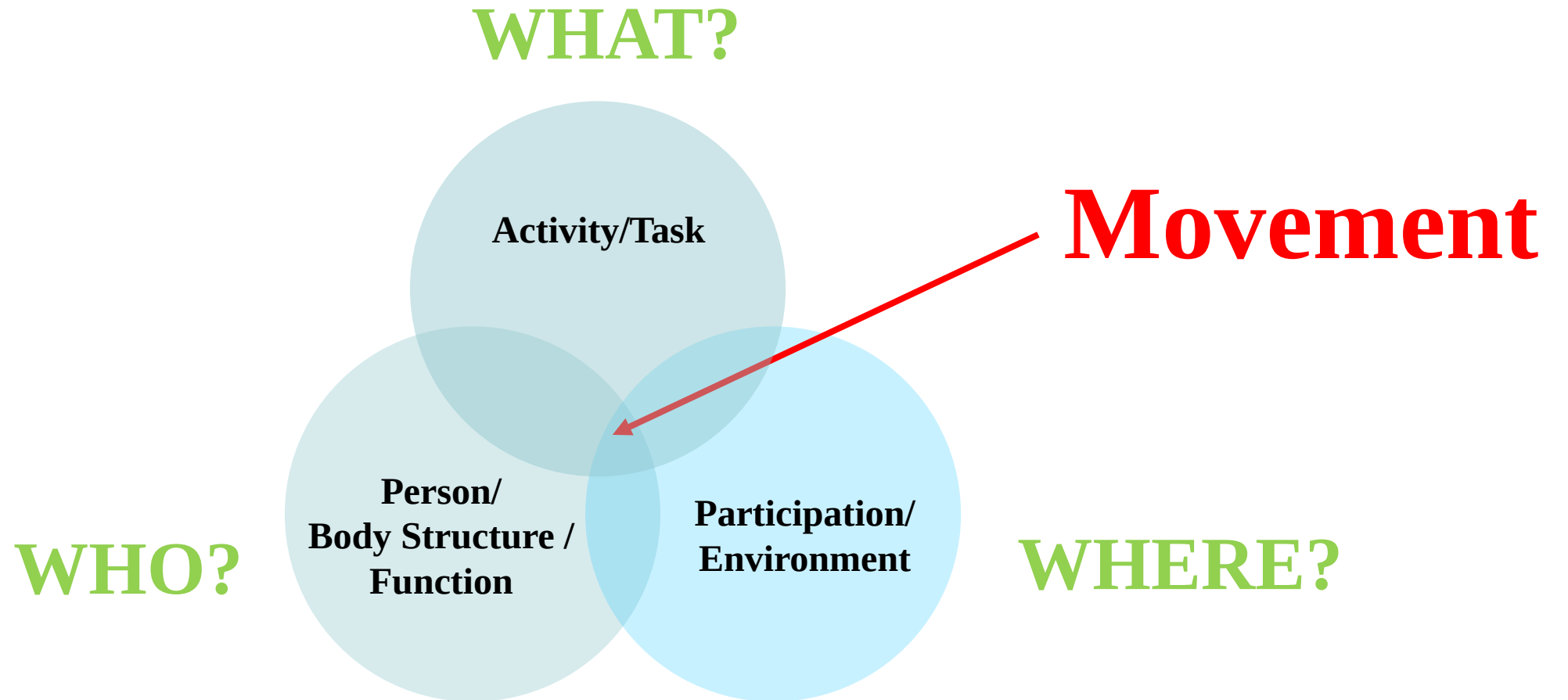


Motor Learning Theory: “Take Home” Message

- **Motor learning probably occurs in stages**
 - Should we give instruction to patients based on activity or based on stage of learning? Or both?
 - Activity of the learner is different in the different stages
- **Activities of the clinician/instructor should be different in the different stages**
 - What will be different in your practice from beginning to advanced stages?

Integrating Motor Control and Learning

Dynamic Interaction of 3 Elements → Movement



The Person Body Structure & Function

Movement System

The Person



ABILITY/ATTRIBUTE	EXAMPLE
Attitudes	Open, closed, or neutral to new experiences
Body type	Stocky, tall, short, lean, muscular, round
Cultural background	Ethnicity, race, religion, socioeconomic
Emotional makeup	Boredom, excitement, fear, joy
Fitness level	Low, moderate, high
Learning style	Visual, verbal, kinesthetic
Maturational level	Immature, intermediate, mature
Motivational level	Low moderate, high
Previous social experiences	One-on-one, small group, large group
Previous movement experiences	Recreational, instructional, competitive

The Task/Activity

The Task

1. What are the component elements of the task?
2. What body movements are required?
3. How is the task paced?
4. Is there a high perceptual or cognitive component to the task?
5. How much precision is needed? Is it gross motor or fine motor?
6. Is object manipulation required?
7. Must the task be performed under differing conditions?
8. How is the task organized?

Skill/Task: Nature of Movement Organization



Discrete

Distinct Beginning
& End

Short duration

Movement prepared
in advanced

Serial

Discrete actions
linked together

Order is important
Longer movement time
Practice in parts

Some prepared in
advanced, others
adjusted during task

Continuous

Repetitive & Rhythmic
Beginning or end
arbitrary or set by

environment
Longer time span
Determined by goal or
person

Ongoing movement
adjusted during task

The Environment

The Environment

- What are the conditions in which the person performs the task/skill?
- What are the regulatory features of the environment?
 - Regulatory:
 - Objects/People: Size, shape, weight, location
 - Support surface
 - Movement: Predictable/Unpredictable
 - Non-Regulatory
 - Background noises and Distractions
- Do the regulatory conditions remain fixed or do they change with each attempt?
 - Inter-Trial Variability:
 - Fixed: Movement is stable and does not require much if any monitoring
 - Changes: Movement diversifies and requires ongoing monitoring

Environmental Predictability

Closed



- Stable and predictable. Allows learner to control the situation. Individuals can pre-plan without fear of environmental change.

Open



- Unpredictable, ever-changing environment. Requires individual to adjust to environment containing objects that have spatial and/or temporal qualities.

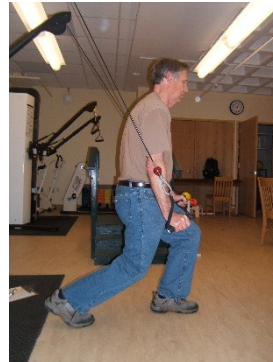
Gentile's Taxonomy

		Task			
Environment		Body Stable No manip	Body Stable Manip	Body xport No Manip	Body xport Manip
	Stationary No Intertrial	Closed Standing Balance	Closed Standing Holding a ball	Closed Walking Hallway	Closed Walking Holding a Ball
	Stationary Intertrial Var	Closed Standing Varied Surfaces	Closed Standing Varied Surface Tossing Ball	Closed Walking Varied Surfaces	Closed Walking Varied Surfaces Tossing Ball
	In-motion No Intertrial	Open Standing Moving Surface	Open Standing Moving Surface Holding Ball	Open Walk up Escalator Holding Ball	Open Walking Tossing Ball (semi quiet hall)
	In-motion Intertrial Var	Open Standing Moving Surface Weight Shifting	Open Standing Moving Surface Tossing Ball	Open Walking Dribbling Ball	Open Dribbling Ball (against Defender)



Stationary
No variability

Stationary
Variability



In Motion
No Variability



In Motion
Variability

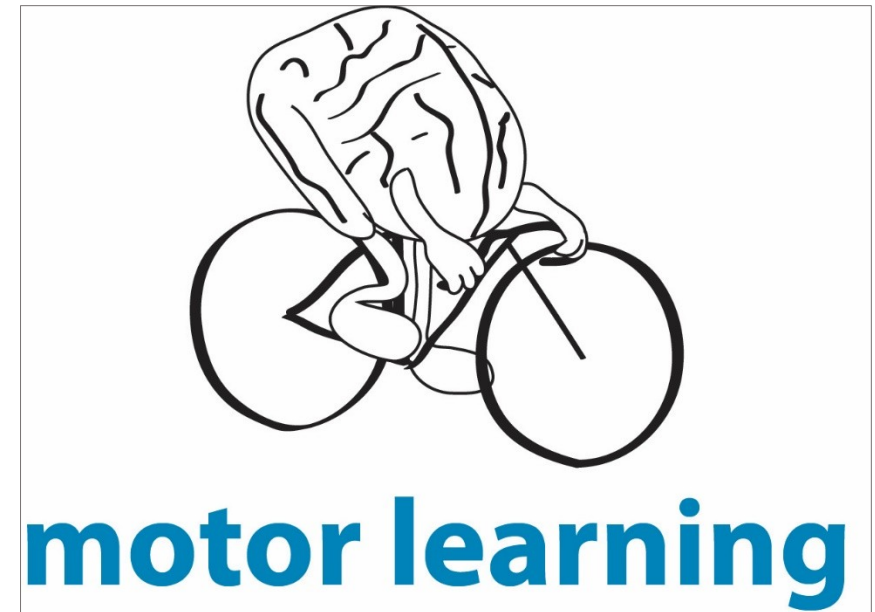


Clinical Implications...

- The stability of the environment may impact the learner differently depending on the stage of learning.
- During early learning, a closed environment maybe beneficial because it requires fewer demands on information processing systems.
- During advanced learning, an open environment maybe beneficial because it provides greater variability.

Steps of Motor Learning

- Present the motor problem to your patient
- Provide opportunity for **PRACTICE**
- Provide **FEEDBACK**
- **EVALUATE** performance, retention, transfer



Assignment one

- Differentiate between motor learning and control.
- Outline the characteristics of motor learning. 10 marks
- Submit by end of day one by 6.00 pm.

End of lesson one